

Code-Layout, Readability and Reusability

Date	7 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Python strictly enforces 4-space indentation; codebase fully adheres to PEP-8 standards.
2	Proper File Structure	Files and folders are logically organized	Yes	Code is cleanly split into src/ (backend logic/DB) and the root directory (UI/configs).
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like emotion_scores, problem_text, and bilstm_result clearly indicate their purpose
4	Function / Method Names	Functions are descriptively named	Yes	Functions like run_prediction(), log_interaction(), and render_analytics() are action-oriented
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings and inline comments explain machine learning processing and UI rendering blocks.
6	Modular Design	Code is split into reusable functions/modules	Yes	Database logic, ML prediction, and the Streamlit UI are isolated in separate .py modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable helper functions were created to prevent duplicating Streamlit UI components.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	try-except blocks wrap the Gemini API calls and DB queries to prevent web server crashes.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	predict.py (Emotion Pipeline)	Python, TensorFlow	Called by the UI whenever the user submits a new study problem.	High
2	database.py (DB Manager)	Python, SQLite	Used by the Auth system, history logger, and analytics dashboard.	High
3	auth.py (Auth System)	Python, bcrypt	Used in the sidebar to handle both Login and Sign-up forms	High
4	Analytics Dashboard UI	Python, Streamlit, Plotly	Renders dynamic charts at the bottom of the main app page.	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Excellent directory structure separating the heavy ML models, the SQLite DB, and the Streamlit UI.
Readability	5	Clean Python syntax with descriptive variable names and a very clear, top-to-bottom UI execution flow.
Reusability	4	Backend logic (like emotion prediction) is completely decoupled from the frontend, making it easy to reuse.
Documentation / Comments	5	All major functions have descriptive docstrings explaining their inputs and outputs.
Overall Score	4.75	Highly maintainable, modular, and professional machine learning codebase.